

Some request commands used for Commissioning and administration have a 'Breadcrumb' argument. When set, this argument SHALL be used to update the value of the [Breadcrumb Attribute](#) as a side-effect of successful execution of those commands. On command failures, the [Breadcrumb Attribute](#) SHALL remain unchanged.

In concurrent connection commissioning flow, the failure of any of the steps [2](#) through [15](#) SHALL result in the Commissioner and Commissionee returning to step [2](#) (device discovery and commissioning channel establishment) and repeating each step. The failure of any of the steps [16](#) through [21](#) in concurrent connection commissioning flow SHALL result in the Commissioner and Commissionee returning to step [16](#) (configuration of operational network information), and MAY result in an attempt to configure secondary network interfaces. In the case of failure of any of the steps [16](#) through [21](#) in concurrent connection commissioning flow, the Commissioner and Commissionee SHALL reuse the existing PASE-derived encryption keys over the commissioning channel and all steps up to and including step [15](#) are considered to have been successfully completed.

In non-concurrent connection commissioning flow, the failure of any of the steps [2](#) through [21](#) SHALL result in the Commissioner and Commissionee returning to step [2](#) (device discovery and commissioning channel establishment) and repeating each step.

Commissioners that need to restart from step [2](#) MAY immediately expire the fail-safe by invoking the [ArmFailSafe command](#) with an [ExpiryLengthSeconds](#) field set to 0. Otherwise, Commissioners will need to wait until the current fail-safe timer has expired for the Commissionee to begin accepting PASE again.

In both concurrent connection commissioning flow and non-concurrent connection commissioning flow, the Commissionee SHALL exit Commissioning Mode after 20 failed attempts.

Once a Commissionee has been successfully commissioned by a Commissioner into its fabric, the commissioned Node SHALL NOT accept any more PASE requests until any one of the following conditions is met:

- Device is factory-reset.
- Device enters commissioning mode.

Ongoing administration of Nodes by Administrators employs many of the same clusters and constraints related to Fail-Safe timer and cluster operation time-outs used for initial or subsequent Commissioning into new Fabrics. The respective cluster specifications for the [Section 11.18, “Operational Credentials Cluster”](#) and the [Section 11.9, “Network Commissioning Cluster”](#) reflect the necessary usage of the [ArmFailSafe](#) and [CommissioningComplete](#) commands of the [Section 11.10, “General Commissioning Cluster”](#) to achieve consistent state during administrative operations.

5.5.2. Commissioning Flow Diagram